Zipf's law

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Problem: Zipf’s law describes a spectacular phenomenon. Where does it come from?

t the 10^{th} most
is 10 times
than the 100^{th}
word, which is
nore frequent
most
which is itself
frequent than
ost frequent
1. Or the 10^{th}
word is twice
than the 20^{th}
word, 4 times

Word frequency ranks can be found in various Websites, such as this one, or this one or this one.

Zipf’s law is an empirical phenomenon. It is not easy to explain.

In 1936, George K. Zipf supposed that the frequency of words in language results from the strategy used by people during their mental search through the vocabulary to find appropriate words to express intended meanings. As a consequence, one may suppose that words are stored by decreasing frequency in our minds.

If so, Kolmogorov complexity may explain Zipf’s law.

- On the one hand, the complexity of words is related to their frequency of use, through the formula: $K(s) \approx \log(1/f) + c$.
- On the other hand, the complexity of words can be inferred from their rank in the vocabulary: $K(s) = \log(1/r) + c$.

This way, Kolmogorov complexity can be seen as acting as a go-between which links rank to frequency.

Explaining Zipf's law

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function as

codes that give us access to meanings.

Consider a meaning m and the word w

that expresses it.

The quantity of information that is

needed to designate m is its complexity

K(m). According to Zipf,

the way we proceed through

Zipf's law

0/1 point (ungraded)

Let's take two books of similar size, written in two different languages (we suppose they have no words in common). Will the meeting of the two books verify Zipf's law?

- ○ yes, because Zipf's law applies to any text
- ○ yes, because both word counts and ranks are affected by a factor two ✓
- ◉ yes, because addition leaves inverse proportionality invariant
- ○ no, because words with the same meaning in the two languages are equally frequent
- ○ no, because word ranks no longer reflects their frequency of use
- ○ no, because addition does not leave inverse proportionality invariant

✗

Answer

Incorrect:

Yes, the reunion of the two foreign books will still verify Zipf's law. This is because word counts are divided by two and ranks are multiplied by two.